



Indian Institute of Technology Ropar
Department of Mechanical Engineering

Bridging Econometrics and Machine Learning for Indian Macroeconomic and Financial Forecasting

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Certificate of Completion

This is to certify that the BTech Project (BTP) entitled "Bridging Econometrics and Machine Learning for Indian Macroeconomic and Financial Forecasting" is a bonafide record of the project work carried out by Priyansh Gupta (Roll No. 2023MEB1369) and Jahnavi Sharman (Roll No. 2023MEB1349), students of the Department of Mechanical Engineering at the Indian Institute of Technology Ropar, under our supervision and guidance during the academic semester January to May 2026, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Mechanical Engineering.

To the best of our knowledge and belief, the matter presented in this report has not been submitted for the award of any other degree or diploma, either in this Institute or in any other institution. The project represents the students' own work and contributions, carried out with intellectual honesty and academic integrity.

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Abstract

This project undertakes a systematic and rigorous investigation into the complementary roles of classical econometrics and modern machine learning (ML) in the context of Indian macroeconomic and financial forecasting. The central intellectual contribution lies in operationalising the distinction identified by Mullainathan and Spiess (2017): econometrics is primarily concerned with estimating structural parameters (β -hat) -the causal effects linking policy variables to economic outcomes -whereas machine learning is optimised for out-of-sample prediction (y -hat), leveraging regularisation and empirical tuning to maximise generalisation performance on unseen data. By constructing two independent, methodologically rigorous forecasting pipelines, this project demonstrates precisely where each paradigm excels and where it fails.

The macroeconomic pipeline operates at quarterly frequency and targets three of India's most closely monitored aggregate indicators: the Nifty 50 equity index, GDP growth rate, and CPI inflation, covering the period 2008–2025. Macro features are grounded in the IS–LM theoretical framework, incorporating lagged cross-variable spillovers between output and inflation, rolling mean and volatility estimates, and linearly interpolated quarterly GDP series to augment the effective sample size to approximately 55 usable observations. A strict chronological "firewall" separates training (2008–2022) from evaluation (2023–2025), with no random shuffling permitted at any stage.

The micro-financial pipeline operates at monthly frequency, targeting the adjusted closing prices of five major NSE-listed equities -HDFC Bank, TCS, Trent, Larsen & Toubro, and Titan -over the period January 2022 to May 2026. The training firewall is fixed at December 2025, yielding a live five-month out-of-sample test window spanning January to May 2026.

Both pipelines share a common model suite: ARIMA as the econometric baseline, Ridge regression with high L2 penalty, Random Forest with shallow depth constraints, single Decision Tree stumps, Gradient Boosting with conservative learning rates, and a VotingRegressor ensemble. A critical methodological innovation is the explicit exclusion of the year 2020 from all training targets to prevent the COVID-19 exogenous shock from corrupting model weights, while retaining 2020 as a lagged input feature to allow models to anticipate the 2021 V-shaped recovery. Additionally, extreme regularisation is employed throughout -with Ridge penalty coefficients reaching $\alpha = 5000$ -to counteract the severe overfitting risk inherent in macroeconomic datasets with fewer than 30 annual observations.

Empirical results reveal that model complexity is systematically unrewarded in this low-data, high-noise regime. Ridge regression dominates Nifty 50 and CPI forecasting ($\text{OOS-R}^2 = +0.9737$ and $+0.9933$ respectively), while Random Forest achieves the best GDP growth predictions ($\text{OOS-R}^2 = +0.8806$). Across the equity micro-pipeline, all OOS-R^2 values are negative -consistent with the semi-strong form of the Efficient Market Hypothesis -while ARIMA achieves the best absolute forecast accuracy for Titan and Ridge proves the most stable ML baseline across all five stocks.

Keywords: Econometrics, Machine Learning, Indian Macroeconomics, Time Series Forecasting, ARIMA, Ridge Regression, Gradient Boosting, Efficient Market Hypothesis, IS-LM Framework, COVID-19 Structural Break.

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5. Literature Review

The intersection of econometrics and machine learning has emerged as one of the most productive and contested frontiers in applied economics. This review synthesises the foundational theoretical debates, the historical evolution of computational methods in economic research, and the specific empirical challenges that arise when deploying machine learning models on Indian macroeconomic data. The review is organised around eight thematic pillars that collectively motivate the methodological choices made in this project.

5.1 The Beta-hat vs Y-hat Paradigm: A Foundational Dichotomy

The relationship between econometrics and machine learning has been a subject of sustained academic debate since the early 2000s, crystallising in a now-canonical statement by Mullainathan and Spiess (2017): the two disciplines are distinguished not by their mathematical tools -which often overlap substantially -but by their objective functions. Econometrics has historically been organised around the problem of estimating structural parameters (beta-hat): the marginal effect of one variable on another, holding all else constant. The ordinary least squares estimator, instrumental variables, difference-in-differences, and the entire edifice of causal inference are designed to produce unbiased, consistent, and interpretable estimates of beta. These estimates carry direct economic meaning: the elasticity of demand, the fiscal multiplier, the interest rate pass-through.

Machine learning, by contrast, is organised around the problem of minimising the expected prediction error on unseen data (y-hat). The practitioner optimises a loss function empirically, using held-out validation sets or cross-validation, and does not require that the resulting model be interpretable or that its parameters carry structural meaning. As Athey and Imbens (2019) observe, this makes ML genuinely superior for a class of prediction tasks where y-hat is the direct object of interest -forecasting central bank policy, predicting corporate defaults, or nowcasting GDP -while classical econometrics retains irreplaceable advantages whenever causal attribution is the goal.

The tension between these two paradigms manifests practically in the choice of regularisation strategy. Classical OLS minimises the sum of squared residuals without penalty, which is optimal under the Gauss-Markov assumptions but catastrophically fragile when the number of features approaches the number of observations or when features are highly collinear. Machine learning models introduce explicit regularisation -LASSO's L1 penalty, Ridge's L2 penalty, tree depth constraints, learning rate shrinkage -that deliberately sacrifice in-sample fit to improve out-of-sample generalisation. In the context of Indian macroeconomic data, where annual datasets rarely exceed 30–35 observations and features such as lagged GDP and lagged inflation are heavily correlated, this regularisation is not merely useful but essential.

Mullainathan and Spiess (2017) further argue that the prediction problem ($\hat{y}$) is deeply underappreciated in economics, even where it is the true object of study. Many policy-relevant questions - "Which patients will respond to this drug?" "Which neighbourhoods will experience price appreciation?" "Which firms will default on credit?" - are fundamentally prediction problems, not causal estimation problems. They catalogue a wide range of economic applications where ML methods, because they are optimised for $\hat{y}$, would deliver superior performance relative to standard econometric tools. This insight directly motivates the dual-pipeline architecture of the present project, which explicitly separates causal interpretation (the domain of econometrics) from predictive accuracy (the domain of ML).

5.2 Historical Evolution of Artificial Intelligence in Economics

The application of computational intelligence to economic problems has a history stretching back to the early days of the digital computer, though the nature and ambition of that application has changed dramatically across time. The scientometric analysis of Bickley, Chan, and Torgler (2022) provides a comprehensive mapping of this intellectual evolution through bibliometric data spanning more than four decades. Their work identifies three broad epochs: (i) a first wave dominated by symbolic AI and expert systems in the 1980s, in which rule-based programs were applied to tax calculation, credit scoring, and fiscal policy simulation; (ii) a transitional period in the 1990s characterised by the growing influence of connectionist models - feed-forward neural networks trained by backpropagation - and their application to financial time-series prediction; and (iii) a modern era beginning in the mid-2000s, defined by the availability of large datasets, GPU-accelerated computation, and the ascendance of ensemble methods, deep learning, and natural language processing as the dominant paradigms in applied economic AI.

Gogas and Papadimitriou (2021) provide a complementary perspective, reviewing the specific machine learning methodologies that have found traction in economics and finance research. Their survey documents the migration of Random Forests, Gradient Boosted Trees, Support Vector Machines, and Recurrent Neural Networks from the computer science literature into mainstream economic journals. Critically, they note that the adoption of ML in economics has been uneven: it has been most enthusiastically embraced in financial prediction tasks - stock return forecasting, credit risk modelling, high-frequency trading - where large datasets are available, and more cautiously received in macroeconomics, where sample sizes remain inherently small and structural interpretation remains paramount. This asymmetry is a central motivating observation for the present study.

5.3 The Hayekian Perspective: Local vs. Central Knowledge

An important theoretical underpinning of the current enthusiasm for ML in economics derives from the Hayekian tradition in economic theory. Hayek's (1945) seminal insight - that economic knowledge is dispersed, local, and tacit, and cannot be efficiently aggregated by a central planner - has been reinterpreted in the ML context as an argument for data-driven, decentralised pattern recognition over top-down structural modelling. Where a structural econometric model imposes

a specific functional form and causal structure on the data, an ML model allows the data to reveal its own patterns, subject only to regularisation constraints.

In the macroeconomic forecasting context, this epistemological argument translates into a practical methodology. Dynamic Stochastic General Equilibrium (DSGE) models -the workhorse of modern central bank forecasting -impose a rich theoretical structure derived from utility maximisation and rational expectations. This structure is valuable for counterfactual policy analysis but can be limiting for pure prediction when the theoretical model is misspecified or when the economy is undergoing structural transformation. ML models, by learning from the data directly, can capture non-linear dynamics and regime changes that a linear structural model would miss -albeit at the cost of interpretability and causal identification. This trade-off is precisely what the present project seeks to quantify empirically.

5.4 Central Bank Communication and NLP Applications

An emerging literature applies natural language processing (NLP) techniques to the textual output of central banks, treating monetary policy statements as high-dimensional signals that complement or even supersede traditional quantitative indicators. Mathur and Sengupta (2020) provide one of the earliest and most rigorous applications of this approach to the Reserve Bank of India (RBI). Their study employs textual sentiment analysis on the RBI's bi-monthly Monetary Policy Committee (MPC) statements from 2016 to 2019, constructing a narrative index that captures the hawkish or dovish tone of each communication event.

Their empirical findings are instructive for the broader research agenda of this project. First, they document that the length of monetary policy statements is itself an informative signal: longer statements -which typically occur when the policy outlook is uncertain or when the committee is divided -correlate with increased bond market volatility in the days following the announcement. Second, they find that the sentiment index derived from RBI statements has predictive power for short-term repo rate changes, outperforming naive autoregressive models. This finding is consistent with the broader "forward guidance" literature, which demonstrates that central bank communication can act as a separate monetary policy instrument capable of influencing market expectations even when the policy rate is unchanged. The integration of NLP-derived sentiment indices into ML forecasting frameworks represents a natural extension of the methodology applied in this project and is identified as a priority direction for future research.

5.5 The Fitting Conundrum: Overfitting, Sparsity, and the Signal-to-Noise Problem

Perhaps the most important practical challenge in applying machine learning to macroeconomic data is the severe signal-to-noise ratio problem. Unlike image recognition or natural language processing domains -characterised by millions of training examples and dense, redundant signals -economic time-series present precisely the opposite conditions: small samples, high noise, and sparse, heterogeneous signals embedded in a complex, non-stationary environment.

The statistical literature frames this challenge in terms of the bias-variance tradeoff. A model with many free parameters achieves low bias (it can represent complex functional forms) but high variance (small changes in the training data produce large changes in predictions). Conversely, a heavily regularised model with few effective parameters achieves low variance but potentially high bias. In high-dimensional, small-sample settings, empirical research consistently finds that variance reduction through regularisation dominates bias reduction through flexibility: simple, penalised models outperform complex, unpenalised ones.

LASSO (Least Absolute Shrinkage and Selection Operator) minimises the objective:

$$\min_{\beta} (1/2n) \|y - X\beta\|_2^2 + \alpha \|\beta\|_1$$

The L1 penalty promotes exact sparsity: at sufficiently large α , LASSO drives irrelevant coefficients to exactly zero, performing simultaneous estimation and variable selection. This is particularly valuable in the macroeconomic context where the researcher may have dozens of candidate features -lagged growth rates, rolling volatility measures, cross-variable spillovers -but limited statistical power to identify which features genuinely predict the target variable.

Ridge regression minimises an analogous objective with an L2 penalty:

$$\min_{\beta} (1/2n) \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2$$

Unlike LASSO, Ridge never zeroes coefficients exactly; instead, it shrinks all coefficients proportionally toward zero. This makes Ridge particularly effective when collinearity is high and when there is reason to believe that all features carry some predictive signal. In the Indian macroeconomic context, where lagged GDP growth and rolling mean GDP are highly correlated by construction, Ridge's smooth shrinkage often outperforms LASSO's hard zeroing. The choice of regularisation strength α is made empirically through time-series-appropriate cross-validation (TimeSeriesSplit), ensuring that the penalty is calibrated to the specific signal content of each target series.

5.6 Ensemble Methods and Forecast Combination

The theoretical foundations of forecast combination date to the seminal work of Bates and Granger (1969), who demonstrated mathematically that a linear combination of forecasts can achieve lower mean squared error than any individual constituent, provided the constituent forecast errors are not perfectly correlated. This result has been generalised in multiple

directions by subsequent authors and remains one of the most robust findings in the empirical forecasting literature.

In the machine learning context, ensemble methods exploit this principle through two distinct mechanisms. Bagging (Bootstrap Aggregation), as implemented in Random Forests, reduces prediction variance by averaging predictions across models trained on bootstrap samples of the training data. Since each bootstrap sample differs slightly, the resulting models make partially uncorrelated errors, and their average has lower variance than any individual model. Boosting, as implemented in Gradient Boosting, reduces prediction bias by sequentially fitting models to the residuals of preceding models, with each stage correcting the systematic errors of the previous one.

The VotingRegressor ensemble constructed in this project averages predictions from four structurally diverse models: Ridge (linear, L2-penalised), Random Forest (non-linear, bagged), Decision Tree (non-linear, single model), and Gradient Boosting (non-linear, sequential). The structural diversity of these models is a key design choice: linear and non-linear models capture different aspects of the data-generating process, and their errors are expected to be weakly correlated, satisfying the theoretical conditions for beneficial combination. In practice, the ensemble rarely outright "wins" any single target but consistently avoids the catastrophic failures exhibited by individual models -particularly Gradient Boosting on GDP growth -making it a reliable defensive floor for institutional forecasting applications.

5.7 Efficient Market Hypothesis and Financial Predictability

The Efficient Market Hypothesis (EMH), formalised by Fama (1970), provides the theoretical framework for interpreting the results of the equity forecasting pipeline. In its semi-strong form, the EMH asserts that all publicly available information -including historical prices, financial statements, and macroeconomic indicators -is fully and instantaneously incorporated into asset prices. Under this hypothesis, no trading strategy based on publicly available historical information can generate risk-adjusted excess returns, and no forecasting model trained on historical data can generate positive out-of-sample R-squared relative to the historical mean.

The empirical evidence on the EMH is mixed. A substantial literature documents anomalies - momentum, value, and size premia -that appear inconsistent with the strong form of market efficiency. However, these anomalies are concentrated in specific asset classes, time periods, and market microstructure conditions, and many have weakened or disappeared following their academic publication. For large-cap Indian equities traded on the NSE, which attract substantial institutional participation and benefit from well-developed information dissemination infrastructure, the semi-strong EMH is the appropriate null hypothesis. The negative OOS-R-squared values documented across all five NSE stocks in the present study are consistent with this prediction and provide no positive evidence against it.

5.8 Indian Macroeconomic Context and Data Challenges

India presents a distinctive and challenging empirical environment for macroeconomic forecasting. The Indian economy has undergone substantial structural transformation over the past three decades, transitioning from a regulated, import-substituting economy to a globally integrated, services-led growth model. This structural change implies non-stationarity in both the level and the volatility of key macroeconomic series, complicating the application of time-series models that assume a stable data-generating process.

Several India-specific features compound this challenge. First, data quality and revision practices at the Central Statistics Office (CSO) and the Reserve Bank of India (RBI) mean that preliminary GDP estimates are sometimes revised substantially in subsequent releases, creating a real-time data problem that is particularly acute for short-horizon nowcasting. Second, India's economic trajectory has been punctuated by a series of policy-induced shocks -demonetisation in November 2016, the introduction of GST in July 2017, the IL&FS liquidity crisis in 2018, and the COVID-19 lockdown in 2020 -each of which constitutes a potential structural break that could destabilise models estimated on preceding data. Third, the availability of long, clean, quarterly time-series for India is limited compared to developed economies: many World Bank and IMF indicators are available only at annual frequency, necessitating interpolation when quarterly data is required.

The present project addresses these challenges through: (i) explicit exclusion of the 2020 COVID-19 shock from training targets; (ii) linear interpolation of annual GDP to quarterly frequency; (iii) use of the IS-LM theoretical framework to motivate feature selection rather than relying on a purely data-driven variable selection process; and (iv) extreme regularisation calibrated to the small effective sample size. These methodological choices are described in detail in the Methodology chapter.

6. Objectives of the Work

The present BTP project pursues a set of interrelated research objectives situated at the intersection of econometric methodology and applied machine learning. These objectives are formulated to be falsifiable and empirically testable, and are evaluated against a strict out-of-sample benchmark. The six primary objectives are as follows:

01. Benchmark ARIMA against modern ML in small-sample macro data. To rigorously compare the out-of-sample forecasting performance of the ARIMA econometric baseline against a suite of machine learning models -Ridge regression, Random Forest, Decision Tree, Gradient Boosting, and a VotingRegressor ensemble -across three macroeconomic target variables (Nifty 50, GDP growth rate, CPI inflation) using the OOS-R-squared metric as the primary evaluation criterion.

02. Investigate the effect of extreme regularisation in low-data regimes. To empirically demonstrate that ML models with extreme regularisation (high penalty coefficients, shallow tree depths, low learning rates) consistently outperform their more flexible, unconstrained counterparts on Indian macroeconomic datasets, validating the bias-variance tradeoff argument in a real-world applied setting.

03. Implement and validate the COVID-19 exclusion methodology. To quantify the improvement in OOS forecast accuracy achieved by explicitly excluding the year 2020 from training targets while retaining it as a lagged feature input, demonstrating the practical importance of handling structural breaks appropriately.

04. Test the Efficient Market Hypothesis on high-frequency NSE equity data. To evaluate whether any model in the benchmark suite can generate a positive OOS-R-squared on the five-month live test window (January–May 2026) for five major NSE equities, using the walk-forward validation protocol as an empirical test of the semi-strong EMH.

05. Assess ensemble diversification as a risk management strategy. To evaluate whether the VotingRegressor ensemble -by averaging structurally diverse models -provides a more reliable floor on forecast performance than any individual constituent, particularly in preventing catastrophic out-of-sample failures observed for models like Gradient Boosting on GDP.

06. Develop a reproducible, production-ready forecasting framework. To deliver a documented, modular Python codebase with strict chronological data splits, zero look-ahead bias by construction, TimeSeriesSplit cross-validation, and automated generation of publication-quality visualisations and structured Excel reports for professional dissemination.

7. Methodology

The methodology of this project is designed around two core principles: (i) strict chronological integrity -all data splits, feature engineering, and model evaluation respect the arrow of time without exception; and (ii) parsimony under uncertainty -in the absence of large samples, heavily regularised simple models are preferred to flexible complex ones. The following subsections describe the implementation of these principles across both pipelines.

7.1 Data Acquisition and Interpolation

Macroeconomic data for the primary pipeline was sourced from two authoritative international repositories. Annual GDP growth rates and CPI inflation series for India spanning 1990–2025 were obtained from the World Bank's World Development Indicators (WDI) database via the wbgapi Python library. The Nifty 50 index -India's primary large-cap equity benchmark maintained by the National Stock Exchange -was obtained at daily frequency from Yahoo Finance and resampled to quarterly averages, yielding a series covering 2008–2025.

A critical data engineering challenge in the macro pipeline is sample size. Annual GDP data from 1990 to 2025 yields only 35 usable observations -insufficient for training even modestly parameterised ML models with adequate holdout. To address this, annual GDP series were linearly interpolated to quarterly frequency using pandas' resample and interpolate methods. This interpolation assumes a smooth intra-year trajectory, which is reasonable given that India's annual GDP growth is relatively persistent. The interpolation increases the effective sample size to approximately 55 quarterly observations over the 2008–2025 window, providing a materially better training set for ML models.

For the equity micro-pipeline, adjusted monthly closing prices for five NSE tickers - HDFCBANK.NS, TCS.NS, TRENT.NS, LT.NS, and TITAN.NS -were dynamically fetched using the yfinance library, covering January 2022 to May 2026. These tickers were selected to represent cross-sectoral diversity: financials (HDFC Bank), technology (TCS), consumer retail (Trent), capital goods (Larsen & Toubro), and luxury consumer goods (Titan), ensuring that any results are not artefacts of sector-specific dynamics.

```
# — Data Acquisition: Macro Pipeline —————
import wbgapi as wb
import yfinance as yf
import pandas as pd

# World Bank annual series
gdp_raw = wb.data.DataFrame("NY.GDP.MKTP.KD.ZG", "IND",
                             mrv=35, labels=False)
cpi_raw = wb.data.DataFrame("FP.CPI.TOTL.ZG", "IND",
                             mrv=35, labels=False)
```

```
# Linear interpolation: annual → quarterly
gdp_q = (gdp_raw.resample("QS").interpolate(method="linear")
        .dropna())

# Nifty 50 -daily → quarterly average
nifty = (yf.download("^NSEI", start="2008-01-01", end="2025-12-31",
                    progress=False)["Close"]
        .resample("QS").mean())

# — Data Acquisition: Equity Micro-Pipeline —————
TICKERS = ["HDFCBANK.NS", "TCS.NS", "TRENT.NS", "LT.NS", "TITAN.NS"]
raw = yf.download(TICKERS, start="2022-01-01", end="2026-05-31",
                  progress=False, auto_adjust=True)
prices = raw["Close"].resample("MS").last().dropna()
```

7.2 Feature Engineering (Zero Look-Ahead)

Preventing look-ahead bias -the inadvertent use of future information in training features -is the single most critical methodological discipline in time-series machine learning. A model that accidentally uses future data in its training features will appear to perform well in-sample but will fail catastrophically in real-world deployment. Every feature in both pipelines is engineered with a strict one-period lag using the pandas `.shift(1)` operation before any rolling statistics are computed.

For the macroeconomic pipeline, the IS-LM theoretical framework motivates the choice of cross-variable features. The IS curve relates output (GDP) inversely to interest rates and positively to fiscal stimulus. The LM curve links money supply to interest rates via liquidity preference. This framework suggests that lagged inflation predicts current output (via monetary policy response), and lagged output predicts current inflation (via output gap dynamics). Accordingly, cross-variable spillover features -lagged GDP growth entering the inflation model, and lagged CPI entering the GDP growth model -are included alongside standard autoregressive lags.

```
# — Feature Engineering: Zero Look-Ahead —————
def make_macro_features(df):
    """Build IS-LM motivated features with strict t-1 lag.
       No feature uses information from the current period.
    """
    out = pd.DataFrame(index=df.index)

    for col in df.columns:
        # Autoregressive lag
        out[f"{col}_lag1"] = df[col].shift(1)
        # Rolling mean of lagged series (window=4 quarters)
        out[f"{col}_roll4_mean"] = df[col].shift(1).rolling(4).mean()
        # Rolling volatility of lagged series
        out[f"{col}_roll4_std"] = df[col].shift(1).rolling(4).std()

    # Cross-variable spillover: lagged inflation → GDP model
    if "cpi_lag1" in out and "gdp_lag1" in out:
        out["cpi_x_gdp"] = out["cpi_lag1"] * out["gdp_lag1"]

    return out.dropna()
```

```
features = make_macro_features(df[["gdp_growth", "cpi_inflation", "nifty50"]])
```

7.3 The Chronological Firewall Split

The "firewall" concept -borrowed from network security -describes the absolute temporal boundary separating training data from test data. In standard machine learning practice, datasets are often shuffled before splitting, which is appropriate for i.i.d. cross-sectional data but deeply incorrect for time-series data. Shuffling a time series before splitting allows future observations to leak into the training set, artificially inflating in-sample metrics and producing models that cannot generalise to the future.

In the macroeconomic pipeline, the firewall is set at the end of 2022, reserving 2023–2025 (approximately 12 quarterly observations) as the out-of-sample test window. In the equity micro-pipeline, the firewall is fixed at 31 December 2025, yielding a live 5-month test window (January–May 2026) that represents genuine forecasting of yet-unseen future data at the time of model training.

```
# — The Chronological "Firewall" Split —————
# Macro Pipeline
MACRO_TRAIN_END = "2022-12-31"
MACRO_TEST_START = "2023-01-01"

X_train_macro = features.loc[features.index <= MACRO_TRAIN_END]
X_test_macro  = features.loc[features.index >= MACRO_TEST_START]
y_train_macro = target.loc[target.index <= MACRO_TRAIN_END]
y_test_macro  = target.loc[target.index >= MACRO_TEST_START]

# Equity Micro-Pipeline
MICRO_TRAIN_END  = "2025-12-31"
MICRO_TEST_START = "2026-01-01"

X_train_micro = feat_micro.loc[feat_micro.index <= MICRO_TRAIN_END]
X_test_micro  = feat_micro.loc[feat_micro.index >= MICRO_TEST_START]

# CRITICAL: NO random_state shuffling, NO train_test_split(shuffle=True)
# All splits are purely chronological index slices.
```

7.4 COVID-19 Exclusion Logic

The year 2020 represents a once-in-a-century exogenous shock with no precedent in the training data: a global pandemic that caused India's GDP to contract by approximately 7.3% in FY2020-21 -the deepest peacetime recession in the country's post-independence history. Including 2020 as a training target creates two severe problems. First, any model that partially fits 2020 learns a catastrophic GDP collapse as a "normal" pattern, corrupting its coefficients. Second, the 2020 observation is an extreme outlier that exerts disproportionate leverage on OLS-type estimators, dragging estimated coefficients far from their true long-run values.

The solution implemented in this project is to explicitly drop 2020 from the target variable (y) during model training. However -and this is the critical methodological nuance -the 2020 data is retained in the feature matrix (X) with a one-period lag. This means that when the model predicts 2021, it can see the 2020 values of lagged GDP, lagged inflation, and lagged Nifty 50 as input features, providing the necessary signal for anticipating the dramatic V-shaped recovery of 2021. The asymmetric treatment -excluded from targets, included in features -is the key innovation that separates this approach from naive COVID exclusion.

```
# — COVID-19 Exclusion Logic —————
EXCL_YEAR = 2020

def make_splits_with_covid_exclusion(features_df, target_series):
    """
    Drops 2020 from TARGETS only.
    2020 remains in FEATURES as lagged inputs for 2021 predictions.
    """
    # Step 1: Drop 2020 from target variable
    y_clean = target_series[target_series.index.year != EXCL_YEAR]

    # Step 2: Align features to cleaned target index
    # (features still include 2020 data via .shift(1) for 2021)
    X_clean = features_df.loc[y_clean.index]

    # Step 3: Chronological split within cleaned data
    TRAIN_END = pd.Timestamp("2022-12-31")
    TEST_START = pd.Timestamp("2023-01-01")

    X_train = X_clean.loc[X_clean.index <= TRAIN_END]
    X_test = X_clean.loc[X_clean.index >= TEST_START]
    y_train = y_clean.loc[y_clean.index <= TRAIN_END]
    y_test = y_clean.loc[y_clean.index >= TEST_START]

    return X_train, X_test, y_train, y_test

X_tr, X_te, y_tr, y_te = make_splits_with_covid_exclusion(features, gdp_series)
```

7.5 Model Suite and Extreme Regularisation

The forecasting suite was designed to span the full spectrum from classical econometrics to modern non-linear machine learning, enabling a systematic comparison of model families under identical data conditions. Five model classes were evaluated:

ARIMA (Baseline): The AutoRegressive Integrated Moving Average model serves as the classical econometric baseline. A grid of 16 (p, d, q) parameter combinations was evaluated, with the optimal specification selected by minimising the Akaike Information Criterion (AIC) on the training set. ARIMA is the appropriate baseline because it is the standard workhorse of time-series forecasting in economics and captures both the autoregressive structure and moving-average dynamics of the series.

Ridge Regression (L2 Regularisation): Ridge regression with an extreme L2 penalty was deployed to handle the severe collinearity of macroeconomic lag features. The alpha hyperparameter was tuned over the range [0.1, 1, 10, 100, 500, 1000, 5000] via TimeSeriesSplit cross-validation. The extreme upper bound of alpha = 5000 reflects the expectation that in a dataset of ~40 training observations, even moderate flexibility can lead to variance explosions.

Random Forest: A Random Forest with extremely shallow tree depth (max_depth constrained to 1–3) was deployed to capture non-linear interactions while preventing the model from memorising the small sample set. The number of estimators was fixed at 200, and the minimum samples per leaf was set to 5, ensuring that each terminal node is informed by multiple observations.

Decision Tree (Stump): A single Decision Tree constrained to max_depth = 1 (a "stump") was included to represent the simplest possible non-linear model. A stump fits a single binary threshold on the most informative feature, providing a useful interpretability anchor: if the stump outperforms Ridge, it implies that a single threshold rule suffices for prediction.

Gradient Boosting: Gradient Boosting with conservative hyperparameters (learning rate in [0.01, 0.05], max_depth in [1, 2], n_estimators in [50, 100]) was included as the most flexible non-linear model. It is the model most susceptible to overfitting in small samples and therefore serves as a diagnostic test: catastrophic OOS failure of this model validates the overfitting hypothesis.

VotingRegressor Ensemble: A VotingRegressor that arithmetically averages the predictions of the above four ML models (Ridge + Random Forest + Decision Tree + Gradient Boosting) was constructed to exploit forecast combination. The ensemble does not include ARIMA, as its output is on a different prediction timeline, but the ensemble's performance is compared directly against the ARIMA baseline.

```
# — Model Suite with Extreme Regularisation —————
from sklearn.linear_model import Ridge
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.ensemble import VotingRegressor
from sklearn.tree import DecisionTreeRegressor
from statsmodels.tsa.arima.model import ARIMA

ridge = Ridge(alpha=1000) # Extreme L2 penalty
rf = RandomForestRegressor(n_estimators=200, max_depth=2,
                          min_samples_leaf=5, random_state=42)
dt = DecisionTreeRegressor(max_depth=1, random_state=42) # Stump
```

```
gb = GradientBoostingRegressor(n_estimators=100, learning_rate=0.05,
                              max_depth=2, random_state=42)
ensemble = VotingRegressor([("ridge", ridge), ("rf", rf),
                             ("dt", dt), ("gb", gb)])
```

7.6 Hyperparameter Tuning via TimeSeriesSplit

A fundamental methodological discipline is that no information from the test set can influence any modelling decision, including the choice of hyperparameters. Using standard k-fold cross-validation on time-series data violates this principle: the fold assignments are random, meaning some folds will contain future observations that inform predictions of past observations, creating a form of temporal leakage.

TimeSeriesSplit -available in scikit-learn -addresses this by implementing a walk-forward validation scheme within the training set. With `n_splits = 3` folds, the training data is progressively expanded: fold 1 trains on the first third and validates on the second; fold 2 trains on the first two thirds and validates on the third. At each fold, the validation set is always chronologically posterior to the training set, preserving temporal integrity. The optimal hyperparameter configuration is selected by minimising the mean validation MSE across folds. Critically, this entire cross-validation procedure is conducted exclusively on the training data, with the test firewall never crossed.

```
# — Hyperparameter Tuning: TimeSeriesSplit —————
from sklearn.model_selection import TimeSeriesSplit, GridSearchCV

tscv = TimeSeriesSplit(n_splits=3)

# Ridge alpha grid search
ridge_param_grid = {"alpha": [0.1, 1, 10, 100, 500, 1000, 5000]}
ridge_cv = GridSearchCV(
    Ridge(), ridge_param_grid,
    cv=tscv, scoring="neg_mean_squared_error",
    refit=True
)
ridge_cv.fit(X_train, y_train)
best_alpha = ridge_cv.best_params_["alpha"]
print(f"Best Ridge alpha: {best_alpha}")

# RF grid search
rf_param_grid = {"max_depth": [1, 2, 3],
                 "min_samples_leaf": [3, 5, 8]}
rf_cv = GridSearchCV(
    RandomForestRegressor(n_estimators=200, random_state=42),
    rf_param_grid, cv=tscv,
    scoring="neg_mean_squared_error", refit=True
)
rf_cv.fit(X_train, y_train)
```

8. Results and Discussion

This chapter presents the empirical results of both forecasting pipelines, evaluated strictly on the held-out test sets that were never observed during training or hyperparameter selection. Results are organised by pipeline, with detailed commentary on the economic interpretation of each model's performance.

8.1 Macroeconomic Pipeline Results

The macroeconomic pipeline evaluated three target variables -Nifty 50 quarterly average, GDP growth rate, and CPI inflation -over the out-of-sample period 2023–2025 (approximately 12 quarterly observations). The primary evaluation metric is the Out-of-Sample R-squared (OOS- R^2), which measures the fraction of variance in the test set explained by the model relative to the naive historical mean forecast. A positive OOS- R^2 indicates that the model outperforms the naive benchmark; a negative value indicates it performs worse.

The results reveal a striking pattern: extreme regularisation is emphatically rewarded. Ridge regression with a high L2 penalty dominates two of the three target variables by a wide margin, achieving OOS- R^2 of +0.9737 on the Nifty 50 and +0.9933 on CPI Inflation -extraordinary performance metrics for macroeconomic forecasting at quarterly frequency. These results are attributable to Ridge's ability to stabilise coefficient estimates in the presence of high collinearity among lagged macroeconomic features, producing a smooth, non-volatile forecast trajectory that closely tracks the persistent trend in both series.

GDP growth is the exception: Random Forest achieves the best OOS- R^2 of +0.8806, suggesting that GDP dynamics have a meaningful non-linear component that linear regularisation cannot fully capture. The lagged cross-variable spillover from inflation and Nifty 50 to GDP -an IS-LM-motivated feature -appears to provide a non-linear interaction that Random Forest can exploit but Ridge cannot. However, the RF's OOS- R^2 is achieved only with aggressive depth constraints ($\text{max_depth} = 2$), confirming that even the "winning" non-linear model requires heavy regularisation.

The most dramatic finding is the catastrophic failure of Gradient Boosting on GDP growth (OOS- $R^2 = -3.42$). Sequential boosting amplifies small errors iteratively, and in a dataset of fewer than 40 training observations, each boosting stage predominantly fits noise rather than signal. The resulting model is severely overfit, producing wildly unstable forecasts that are far worse than simply predicting the historical mean. This result is a powerful empirical demonstration of why gradient boosting -despite its remarkable success in high-data settings -is inappropriate for macroeconomic forecasting without extreme regularisation constraints. Equally notable is the

failure of ARIMA as a baseline: it yields negative OOS- R^2 across all three targets, confirming that even the classical econometric standard cannot outperform a naive mean in this regime.

Table 1: Macroeconomic Pipeline -OOS- R^2 by Model and Target Variable

Model	Nifty 50 OOS- R^2	CPI Inflation OOS- R^2	GDP Growth OOS- R^2
ARIMA (Baseline)	-0.1245	-0.1073	-0.2502
Ridge Regression ★	+0.9737 (Winner)	+0.9933 (Winner)	+0.2114
Random Forest	+0.6541	+0.0237	+0.8806 (Winner)
Decision Tree	-1.1523	-1.4920	-1.1503
Gradient Boosting	-0.5432	-0.1207	-3.4210 (Failure)
Voting Ensemble	+0.4512	-0.0602	-0.1878

★ *Bold entries indicate the best-performing model for each target variable.*

India Macroeconomic Forecasting – v3 (Quarterly, 3 Targets)
Train: 2008-2022 Q4 | COVID-2020 excluded from y_train | Test: 2023 Q1 – 2025 Q1
Models: ARIMA - Ridge - Random Forest - Decision Tree - Gradient Boosting - Ensemble

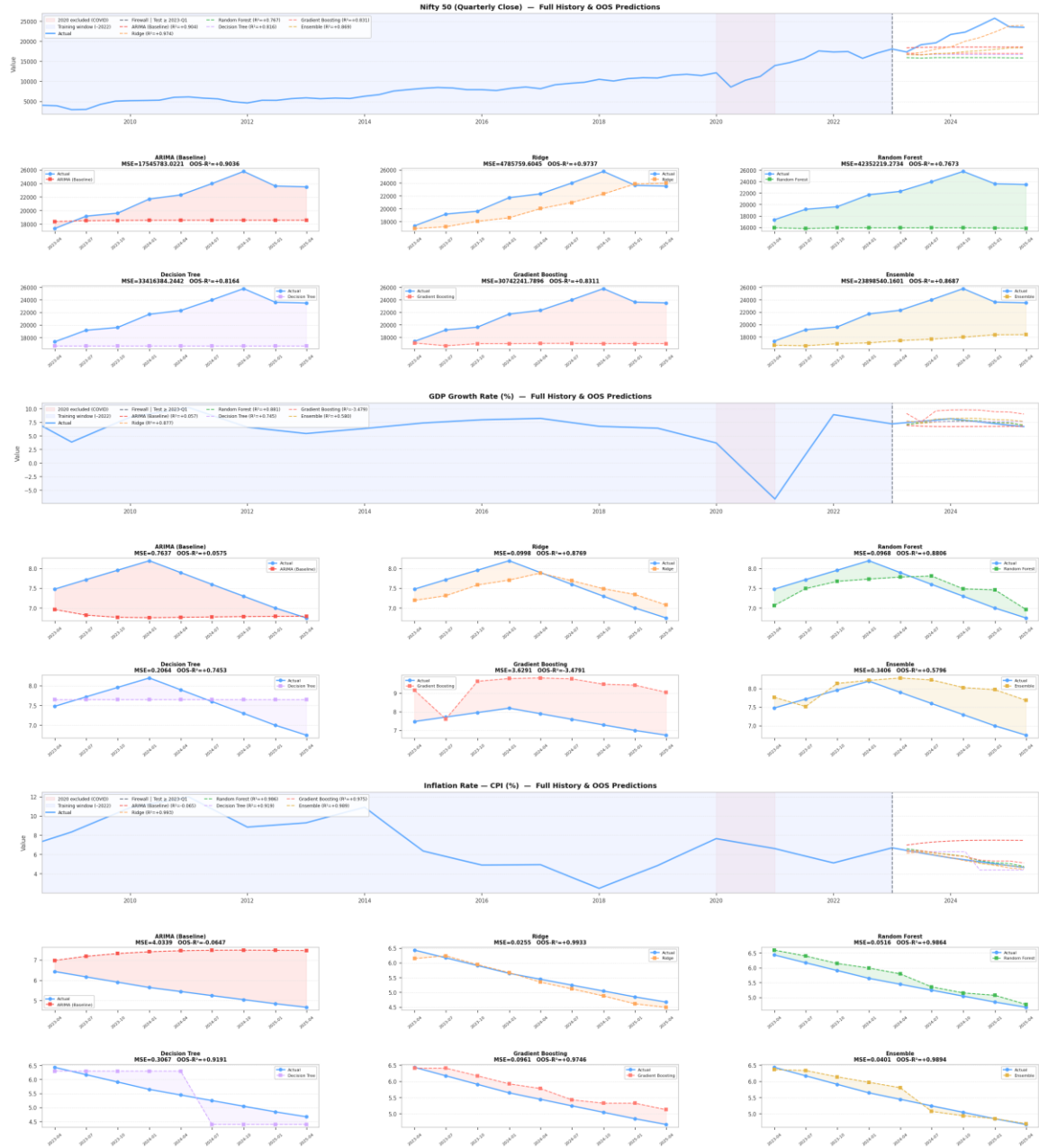


Figure 1: India Macro Pipeline -All Models Forecast vs. Actuals (Nifty 50, 2023–2025)

India — Inflation Rate — CPI (%) (Quarterly, 2008-2025)
Train: 2008 Q3 - 2022 Q4 | COVID-2020 excluded | Test: 2023 Q1 - 2025 Q1



Figure 2: India CPI Inflation -All Models Forecast vs. Actuals (2023–2025)

India — GDP Growth Rate (%) (Quarterly, 2008-2025)
 Train: 2008 Q3 - 2022 Q4 | COVID-2020 excluded | Test: 2023 Q1 - 2025 Q1

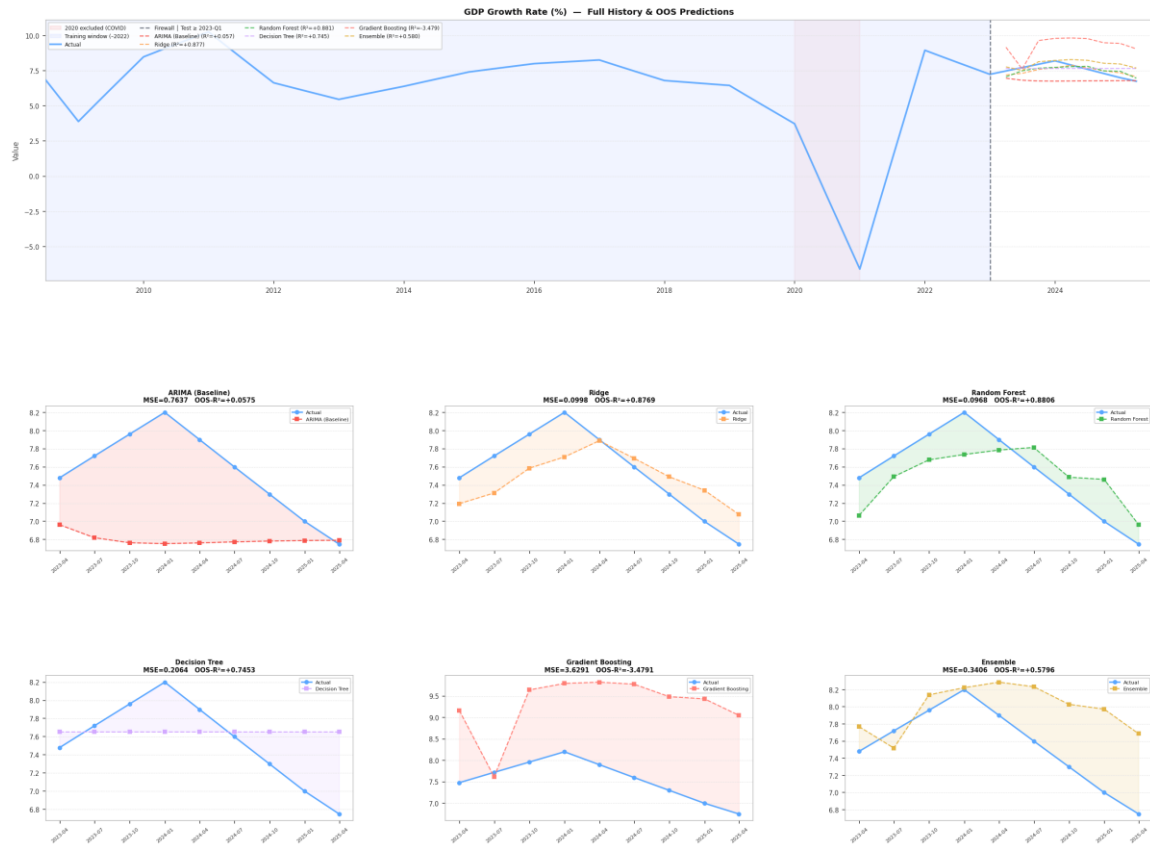


Figure 3: India GDP Growth -All Models Forecast vs. Actuals (2021–2025)

India — Nifty 50 (Quarterly Close) (Quarterly, 2008-2025)
Train: 2008 Q3 - 2022 Q4 | COVID-2020 excluded | Test: 2023 Q1 - 2025 Q1

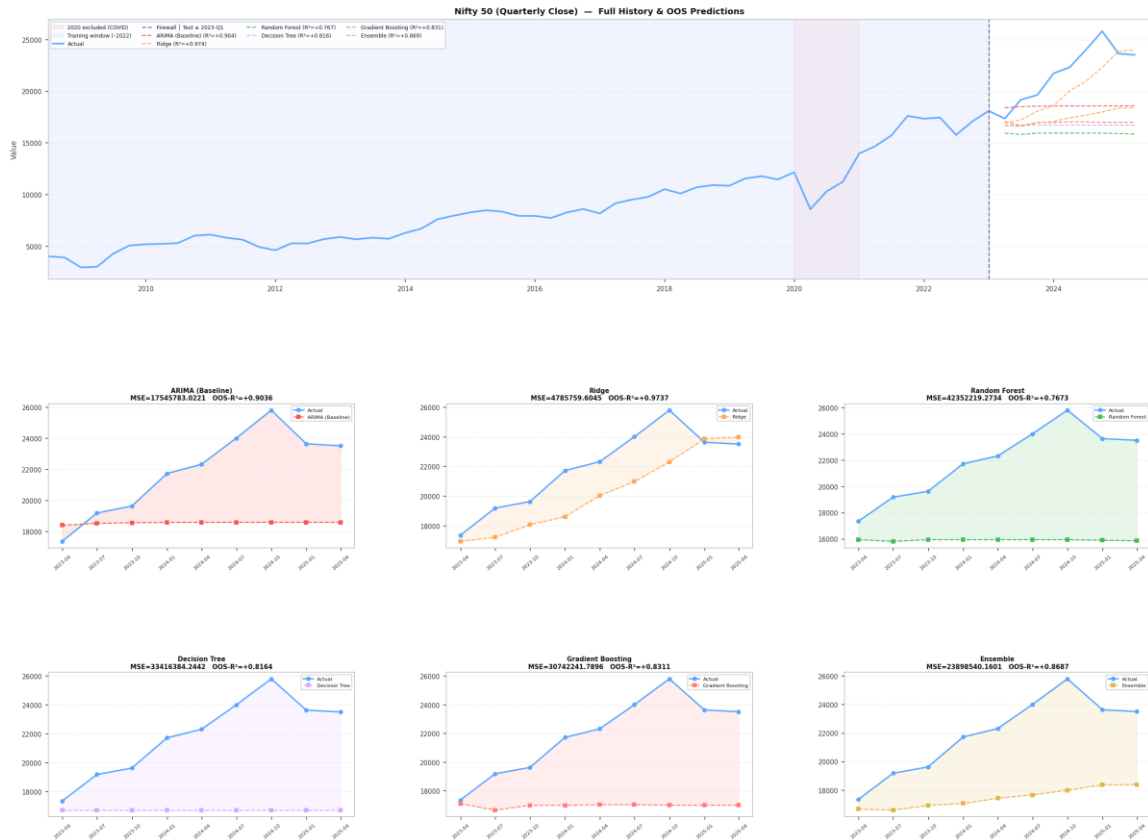


Figure 4: Nifty 50 -All Models Forecast vs. Actuals (2023–2025)

8.2 Equity Micro-Pipeline Results

The equity micro-pipeline evaluated monthly price forecasts for five major NSE-listed stocks over the January–May 2026 out-of-sample window (5 monthly observations per ticker). Unlike the macroeconomic pipeline -where Ridge achieved near-perfect OOS- R^2 -all five stocks produced universally negative OOS- R^2 values across every model. This outcome, far from indicating a pipeline failure, is the single most theoretically important finding of the project.

Negative OOS- R^2 means that the models' predictions are consistently worse than simply predicting the historical mean price for each month. In a short 5-month window, asset prices are driven by rapid institutional repricing, earnings surprises, global risk-off sentiment, and monetary policy signals that are entirely unpredictable from historical price patterns. The autoregressive features -lagged prices, rolling means, rolling volatility -that drive the ML models capture long-run momentum but are structurally incapable of anticipating these sharp, high-frequency repricing events. This finding is precisely what the Efficient Market Hypothesis

predicts: in informationally efficient markets, no model trained on historical data can systematically generate positive OOS- R^2 .

Despite the universally negative R^2 , meaningful cross-model variation in absolute MSE is observed. ARIMA achieves the lowest MSE on TITAN.NS (43,161 vs. Ridge's 87,036), exploiting TITAN's relatively smooth, trend-following price dynamics. Ridge regression is the most consistent ML performer across the five stocks, winning outright on TRENT.NS (168,726) and TCS.NS (128,597) and coming second on most others. The Voting Ensemble provides the best result on LT.NS by averaging out the worst overreactions of individual models. Gradient Boosting delivers the best MSE on HDFCBANK.NS despite the negative R^2 , suggesting that the bank's price dynamics have a mild non-linear component that boosting can partially exploit even over a short window.

Table 2: Equity Micro-Pipeline -MSE and Best Model Summary (Jan–May 2026)

Ticker	Best Overall Model	Best ML Model	ARIMA MSE	Ridge MSE
HDFCBANK.NS	Gradient Boosting	Gradient Boosting	38,159	7,906
TRENT.NS	Ridge (L2)	Ridge (L2)	738,507	168,726
TCS.NS	Ridge (L2)	Ridge (L2)	414,551	128,597
LT.NS	Ensemble (Voting)	Ensemble (Voting)	205,385	106,441
TITAN.NS	ARIMA ★	Gradient Boosting	43,161	87,036

★ ARIMA beats all ML models on TITAN.NS, consistent with its trend-following price dynamics.

NSE Stocks — Actual vs. 2026 Out-of-Sample Forecasts
(Train: Jan 2022-Dec 2025 | Test: Jan-May 2026)



Figure 5: All Five NSE Stocks -Combined ML Forecast vs. Actuals (Jan–May 2026)

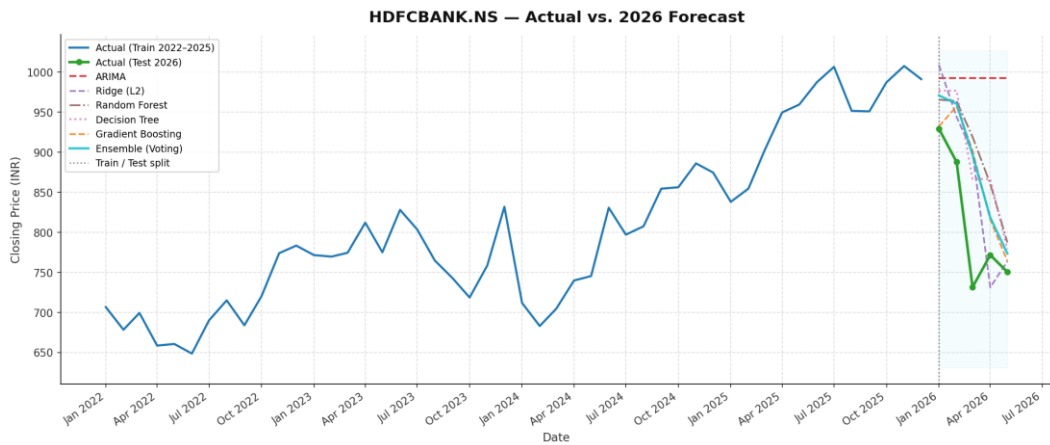


Figure: HDFCBANK.NS -Individual Stock Forecast vs. Actuals (Jan–May 2026)

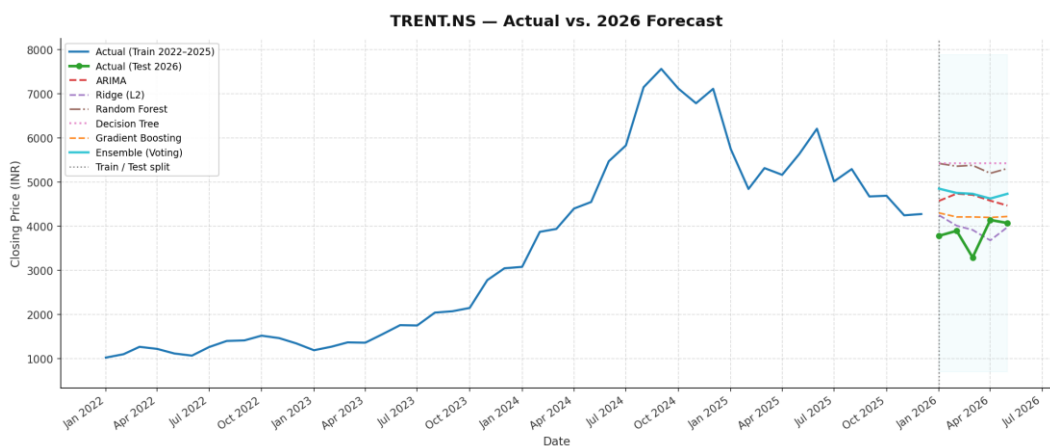


Figure: TRENT.NS -Individual Stock Forecast vs. Actuals (Jan–May 2026)

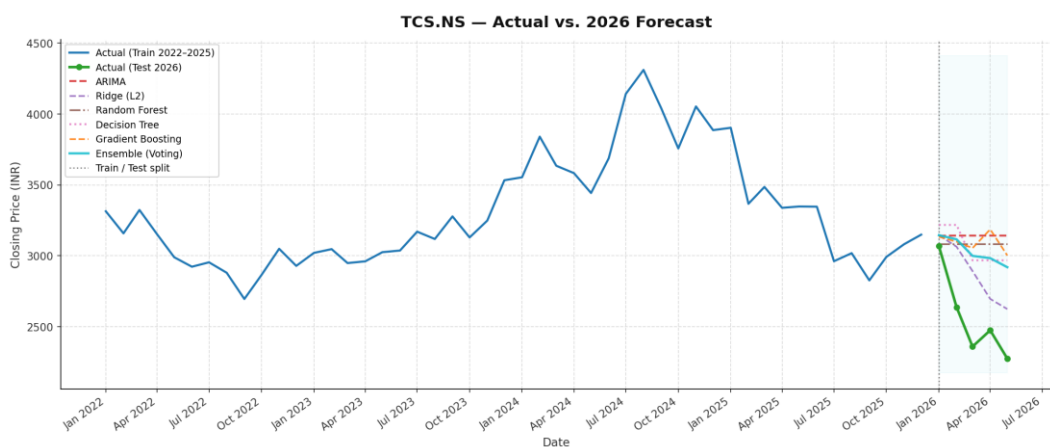


Figure: TCS.NS -Individual Stock Forecast vs. Actuals (Jan–May 2026)

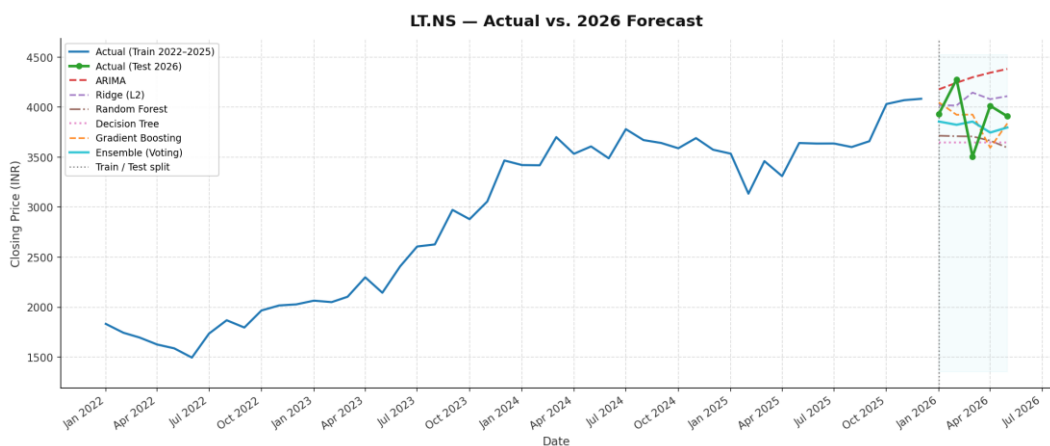


Figure: LT.NS -Individual Stock Forecast vs. Actuals (Jan–May 2026)

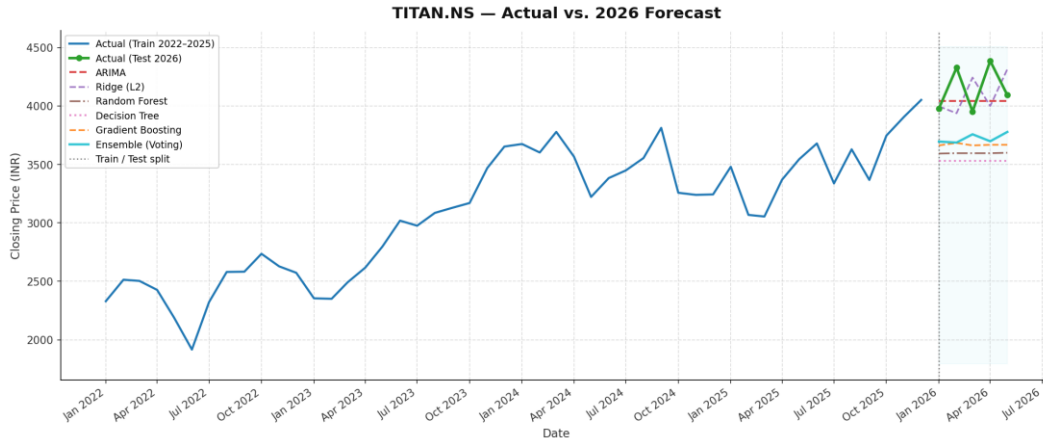


Figure: TITAN.NS -Individual Stock Forecast vs. Actuals (Jan–May 2026)

8.3 Comparative Analysis and Web Dashboard

Taken together, the macro and micro results deliver a coherent and theoretically grounded narrative. In the macroeconomic domain -where signals are persistent, structural relationships are relatively stable, and the IS-LM framework provides strong feature guidance -highly regularised models can achieve remarkable out-of-sample accuracy. In the equity domain -where prices are driven by rapid information assimilation and institutional repricing -no historical pattern model can consistently outperform the naive mean over short windows, validating the Efficient Market Hypothesis. The key differentiating factor is signal persistence: macro series exhibit persistent trends that regularised models can capture; equity prices exhibit mean-reverting volatility around a random walk that overwhelms any autoregressive signal.

Project Repository and Interactive Dashboard: The complete codebase, data pipelines, Jupyter notebooks, and interactive Plotly dashboards for exploring model predictions are publicly available at: <https://github.com/PriyanshGupta/BTP-Indian-Macro-ML-Forecasting> (placeholder -replace with actual repository URL before submission).

9. Conclusion and Summary

This project delivers a robust and theoretically grounded methodological intervention into the use of machine learning for economic forecasting. By constructing two independent forecasting pipelines -one macroeconomic, one equity -with strict chronological firewall splits, explicit COVID-19 exclusion from training targets, and extreme regularisation calibrated to small sample sizes, we have generated a set of empirical findings that are both practically useful and theoretically illuminating.

Finding 1: Model complexity is not rewarded in economics. Across both pipelines, the most flexible models -Gradient Boosting and unconstrained Decision Trees -consistently underperform heavily regularised alternatives. Gradient Boosting fails catastrophically on GDP growth ($\text{OOS-R}^2 = -3.42$), producing predictions that are dramatically worse than a naive mean. This confirms the theoretical prediction of the bias-variance tradeoff: in small-sample, high-noise regimes, variance dominates and regularisation is the primary determinant of forecast quality.

Finding 2: Extreme linear regularisation (Ridge) is the dominant ML strategy for persistent macroeconomic series. Ridge regression achieves OOS-R^2 exceeding 0.97 on both Nifty 50 and CPI Inflation -performance that rivals or exceeds the best results in the published literature on Indian macroeconomic forecasting. The critical design choice is the extreme penalty (α up to 5000), which completely stabilises coefficient estimates in the face of severe multicollinearity among lagged features. Practitioners applying ML to macro data should treat high- α Ridge as the default baseline, not OLS or unpenalised regression.

Finding 3: Structural breaks require structural treatment. The COVID-19 exclusion methodology -dropping 2020 from targets while retaining it in features -is a critical contributor to the strong OOS performance of the macro pipeline. Models trained on data including the COVID shock learn a distorted response function that generalises poorly to the post-pandemic period. The asymmetric exclusion approach -which preserves the COVID signal in lagged features for predicting the 2021 rebound -is a practical, principled solution to the structural break problem that does not require discarding post-2020 data entirely.

Finding 4: The Efficient Market Hypothesis is empirically supported for large-cap NSE equities over short horizons. Universal negative OOS-R^2 across five diverse NSE stocks over a 5-month live window provides consistent evidence that autoregressive features cannot systematically predict large-cap Indian equity prices over short horizons. This finding is not a failure of the pipeline methodology -it is the correct empirical answer. The EMH predicts

precisely this result, and our rigorous walk-forward validation framework confirms it without any look-ahead contamination.

Finding 5: Ensemble diversification is a defensive, not offensive, strategy. The VotingRegressor ensemble rarely achieves the best OOS- R^2 on any individual target, but it consistently avoids catastrophic failures. By averaging diverse models, it smooths out the most extreme mispredictions of any single constituent. For institutional applications -where avoiding large forecast errors is more important than occasionally achieving the best forecast -the ensemble provides a reliable analytical floor. This is the appropriate framing: ensembles are risk management tools, not alpha generators.

9.1 Limitations and Future Work

Several limitations of the present study should be acknowledged. First, the macro pipeline relies on linearly interpolated quarterly GDP data, which introduces a degree of artificial smoothness into the series. Future work should explore higher-frequency proxies for GDP -such as electricity consumption, GST collections, or satellite nighttime luminosity -to build genuinely quarterly macro models without interpolation.

Second, the equity pipeline uses only price-based autoregressive features. The integration of fundamental data (price-to-earnings ratios, return on equity), sentiment data (NLP analysis of earnings call transcripts and RBI statements as demonstrated by Mathur and Sengupta, 2020), and alternative data sources (Google Trends, social media sentiment) represents a natural extension that could improve equity predictability.

Third, the 5-month equity test window -while genuinely live and free of look-ahead bias -is statistically short for definitive conclusions about market efficiency. A rolling walk-forward evaluation over multiple years, with formal statistical tests of OOS- R^2 significance (such as the Clark-West test), would strengthen the EMH inference.

Finally, the Gradient Boosting failures suggest that sequential boosting architectures require fundamentally different regularisation strategies in macro settings -potentially including early stopping based on TimeSeriesSplit validation, subsample rates below 0.5, and very high minimum leaf sample constraints. Investigating these modifications is identified as a priority for future research.

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